

Monitoring & Observability in Cloud AIoT



Monitoring

Why Continuous Monitoring Matters

- AIoT systems operate 24×7 and require ongoing visibility.

Monitoring & Observability

- Ensures reliability & performance
- Builds trust in AI-driven decisions
- Enables proactive issue resolution



Observability

Monitoring Tracks

- 📶 Device health & connectivity
- 🔄 Data flow & pipeline performance
- 📱 AI model accuracy & drift
- ⚙️ System uptime & reliability

Observability: Going Beyond Monitoring

- 🔍 Understand *why* systems behave a certain way
- 📊 Correlate logs, metrics & traces
- ⚠️ Identify root causes faster

Observability transforms data into operational confidence

Key Components of Cloud AIoT Observability



Logs

- Detailed records of events and application behavior.



Metrics

- Numerical data representing system health over time (e.g., CPU, memory usage).



Traces

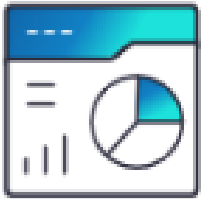
- End-to-end paths of requests across distributed microservices.



AI-Driven Insights

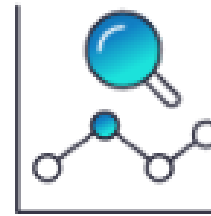
- Machine learning predicts resource needs (CPU/memory) and analyzes data for anomalies, enabling automated resource allocation.

Monitoring vs. Observability in AIoT



Monitoring (Reactive)

- 🔔 Alerts when predefined thresholds are crossed
- 📶 Detects device offline status or connectivity loss
- ⚠️ Triggers alarms for CPU spikes or service failures



Observability (Proactive)

- 🔍 Explains *why* issues occur
- 🤖 Investigates incorrect AI predictions or latency
- 📊 Correlates logs, metrics & traces for root cause analysis

Monitoring tells you **what happened**. Observability explains **why it happened**.
Together they ensure reliability, transparency, and trust in AIoT systems

Benefits and Best Practices



Enhanced Reliability

- Detect issues before they impact users
- Prevent downtime & service disruptions



Operational Efficiency

- Link insights from edge devices to cloud services
- Accelerate troubleshooting & root cause analysis

Best Practices



Define Clear KPIs

Monitor uptime, latency, model accuracy & device health



Use Unified & Automated Tools

Integrate logs, metrics & traces in one platform



Enable Actionable Alerts

Automate alerts & responses to critical events

Proactive visibility ensures resilient and efficient AIoT operations

Popular Tools for AIoT Monitoring & Observability

Open-Source Monitoring & Visualization



Prometheus

✓ Collects & stores time-series metrics



Grafana

✓ Visualizes metrics & real-time dashboards

Cloud-Native Monitoring Services



Google Cloud Monitoring

✓ Unified monitoring & alerting for cloud resources



AWS CloudWatch

✓ Metrics, logs & automated alerts for AWS workloads

Enterprise Observability Platforms



Splunk Observability Cloud

✓ Advanced analytics & full-stack observability
✓ AI-driven insights & anomaly detection

Combine monitoring, visualization, and analytics for complete system visibility